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Exploring Agility in Distributed Information Systems Development Teams: An Interpretive Study in an Offshoring Context

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Agility is increasingly being seen as an essential element underlying the effectiveness of globally distributed information systems development (ISD) teams today. However, for a variety of reasons, such teams are often unable to develop and enact agility in dealing with changing situations. This paper seeks to provide a deeper understanding of agility through an intensive study of the distributed ISD experience in TECHCOM, an organization widely recognized for its excellence in IT development and use. The study reveals that agility should be viewed as a multifaceted concept having three dimensions: resource, process, and linkage. Resource agility is based on the distributed development team's access to necessary human and technological resources. Process agility pertains to the agility that originates in the team's systems development method guiding the project, its environmental scanning, and sense-making routines to anticipate possible crises, and its work practices enabling collaboration across time zones. Linkage agility arises from the nature of interactional relationships within the distributed team and with relevant project stakeholders, and is composed of cultural and communicative elements. The paper highlights some of the difficulties in developing agility in distributed ISD settings, provides actionable tactics, and suggests contingencies wherein different facets of agility may become more (or less) critical.

Key words: agility; global software development; distributed information systems development; virtual teamwork; offshoring; insourcing; interpretive study; case study

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1. Introduction

Today, globally distributed information systems development (ISD) is being widely used in organizations. A number of presumed advantages such as access to a large pool of qualified human resources or "intellectual horsepower" (Holstrom et al. 2006, p. 12), significant cost advantages, speeding up of work by "following the sun" (e.g., Carmel 1999, Sarker and Sahay 2004), and enhanced "strategic flexibility" (Breu and Hemingway 2004) have prompted over 95% of *Fortune* 1,000 firms to develop some of their systems in such settings (e.g., Robinson and Kalakota 2004). Not surprisingly, distributed ISD in a global context is increasingly capturing the attention

of both researchers and practitioners, with several issues such as communication and coordination (e.g., Sarker and Sahay 2003), appropriate ISD techniques (e.g., Kotlarsky et al. 2007), decision and negotiation processes (e.g., Mohan et al. 2006), among others, being investigated and discussed in the literature. With the passage of time, extensive discussions about the challenges faced in the practice of distributed ISD, especially those related to the geographical, temporal, and cultural distances (e.g., Holmstrom et al. 2006, Sarker and Sahay 2004, Carmel 1999), have also emerged.

A troubling point emerging in the literature is that distributed ISD team structures, although

apparently directed toward enhancing the *strategic flexibility* of organizations, are in many respects quite *rigid* (Breu and Hemingway 2004). In an age that demands “faster-cycle technological innovation,” organizational team configurations need to develop nimbleness (Lui and Piccoli 2007, p. 122). Furthermore, because globally distributed ISD teams are increasingly being managed with a strategic focus rather than a cost focus (e.g., Carmel and Agarwal 2002), it becomes critical for such teams to develop *agility* (Fitzgerald et al. 2006) to enhance their effectiveness (e.g., Lui and Piccoli 2007, van Oosterhout et al. 2006).

The concept of agility is rooted in agile manufacturing (Yusuf et al. 1999) and has influenced several other disciplines, such as supply chain management and organizational management (e.g., Christopher 2000), fields that require alignment amongst different elements and need to react to a fast-changing environment. However, it is only recently that the concept of agility has gained the attention of IS researchers and practitioners, particularly in the context of ISD. Note that the literature has generally focused on examining the use (and customization) of agile methods in ISD projects (e.g., Balasubramaniam et al. 2006, Fitzgerald et al. 2006, Flor 2006, Holmstrom et al. 2006). Although recognizing the value of this body of work, some scholars (e.g., Conboy 2009) have suggested that there is confusion surrounding this concept in the IS and ISD arenas. Notably, van Oosterhout et al. (2006, p. 133) argue that although “much has been said and written on” agility, a “consensus on a definition of agility has not yet emerged.” Along similar lines, Börjesson and Mathiassen (2005) assert that there is little research on understanding what the relevant agility principles or approaches are for ISD. Furthermore, Conboy and Fitzgerald (2007) highlight the lack of rigor and cohesion in the manner in which the agility concept is used in ISD.

The confusion surrounding agility is arguably even greater in the globally distributed team arena. Some researchers, for example, have argued that the fundamental concepts underlying agility, such as adaptive product development, and close/constant interaction between team members, are difficult to achieve in a *distributed* ISD setting (e.g., Balasubramaniam et al. 2006, Mistry and Navelkar 2006). The problems of

developing and maintaining agility in distributed structures have been echoed in manufacturing as well (Liu and Sycara 1997, Xu et al. 2005), the discipline within which the concept of agility has reached a high level of maturity. On a similar note, based on his “practical experience in globally distributed teams,” Simons (2006) concluded that expecting agility is “impractical and inefficient” in such settings. We thus believe that, given the widespread occurrence of globally distributed ISD, it is important to clarify our collective understanding of agility for such settings, and to also examine the mechanisms that might lead to higher levels of agility. The objective of this paper, then, is to contribute toward this goal, specifically *by developing an empirically grounded framework of agility in distributed ISD settings and by identifying some actionable tactics for enhancing agility within such settings*. We seek to do this through an intensive study of a U.S.-based organization, which specifically engages in globally distributed ISD projects through its offshore insourcing initiatives. In developing our framework, we also draw inspiration from research on agility in diverse areas, especially, manufacturing, ISD, supply chain management, and marketing, all of which represent areas that are susceptible to changing needs/demands. Finally, in an effort to provide more actionable guidance to practitioners in terms of deciding which aspect of agility (and associated tactics) to use, given certain circumstances, we also identify some of the *contingencies that may influence the importance of the agility facets*.

The rest of the manuscript is organized as follows: First, we briefly review the state of research on agility, followed by a discussion of our research methodology. Next, we elaborate on the conceptual framework of agility in distributed teams that emerged from our interpretation of the data. Thereafter, we identify a preliminary set of contingencies that may impact which aspect of agility tends to become more salient for a given distributed ISD team. Finally, we conclude with a discussion of key contributions.

2. A Brief Review of Agility

The concept of agility has its roots in agile manufacturing (Meredith and Francis 2000), and was initiated and “popularized” by a group of researchers at the Iaccoca Institute of Lehigh University in 1991 (Yusuf

et al. 1999, p. 33). Although agility in manufacturing draws on tenets of “lean manufacturing,” which owes its origin to the Toyota Production System (Ohno 1988), scholars warn that agility should “not be confused with leanness.” Leanness means “containing little fat,” whereas agility implies “nimbleness” (Christopher 2000, p. 38). Within manufacturing, agility has also been viewed as speedy, “innovation proactivity, quality, and profitability” using integrated and “reconfigurable resources and best practices” (Yusuf et al. 1999, p. 37), and attempting to deliver “new products efficiently to the customer-driven market” (Zhou and Nagi 2002, p. 14). The concept of agility has been applied to supply chains as well, referring to their market sensitivity, process integrativeness, and virtuality (Christopher 2000). Within the IS discipline, Sambamurthy et al. (2003) and Overby et al. (2006) have conceptualized agility as incorporating the “sensing and response capabilities” of the firm. Similarly, Fink and Neumann (2007) define IT-dependent organizational agility as the “ability to respond operationally and strategically to changes in the external environment through IT.”

In the context of ISD, agility has been seen as the ability of “information systems development and deployment methods to swiftly adapt to the changing business requirements” (Lee et al. 2006, p. 50). Adopting a more focused perspective, researchers in this area (e.g., Conboy and Fitzgerald 2007) suggest that ISD agility often arises from the use of agile methods (e.g., Scrum, XP) and practices, such that organizations and ISD project teams can handle change quickly. Some researchers, however, have suggested that there are *other potent bases* of agility for distributed ISD teams (in addition to agile methods) that need to be investigated (e.g., Lee et al. 2006, Conboy 2009).

A major theme that emerges from review of the literature is that agility is *multidimensional*. For example, in the manufacturing and supply chain literature, Meredith and Francis (2000) have developed the “agile wheel” composed of four different quadrants of agility: strategic agility, process agility, linkage agility, and people agility. On similar lines, Lui and Piccoli (2007, pp. 125–126) have argued that information systems agility is composed of technological agility, process agility, people agility, and structure agility. Given

the complexities involved in globally distributed ISD, we can expect a multifaceted view of agility within the distributed ISD context as well, which needs to be uncovered; scholars such as Miller and Friesen (1984) offer useful advice in this regard. Liking a complex (multidimensional) construct to a painting with various underlying elements, Miller and Friesen (1984, p. 29) suggest that to understand the painting, the audience first analyzes its primary components or attributes such as the brushstrokes, colors, etc., and then considers “their role in making up that [the] whole” painting (Miller and Friesen 1984, p. 29). In other words, the first step for researchers studying a multidimensional concept would be to identify its different elements, followed by an investigation of how these different “elements interrelate and complement each other” (Miller 1996, p. 7). Consistent with Miller and Friesen’s view (1984), our first (and primary) step involves the identification of the core dimensions/subdimensions (“elements”) of agility in distributed ISD teams; as a next step, informed by the contingency perspective, we suggest how these elements may be interrelated for different distributed ISD settings.

3. Research Methodology

3.1. Research Context

Our study was conducted in a multinational high-tech organization, which we refer to as TECHCOM (a pseudonym). The division of TECHCOM that we focused on (*B2B division*, a pseudonym) is responsible for developing and maintaining systems that enable the company’s relationships with large buyers and suppliers. The division has IT-related personnel in two cities in the United States (*USCity1*—the headquarters, with managers, and developers, and *USCity2*, with managers and production support personnel), Europe (*EuropeCity1*, with developers, testers, and production support personnel), China (*ChinaCity1*, with developers), and India (*IndiaCity1*, with managers, developers, and testers).¹ Although TECHCOM primarily uses offshore *insourcing* to countries such as India through its captive centres to

¹ We do not name the specific cities to ensure that the organization’s identity is not revealed.

gain productivity and cost advantages, it also engages in *offshore outsourcing*. That is, it uses the services of external IT vendors at its offshore sites, including *IndiaCity1*, if and when necessary. Thus, distributed projects often have TECHCOM employees from different locations (say, *USCity1*, *USCity2*, *ChinaCity1*, and *IndiaCity1*) blended with employees from one or more IT vendors.

3.2. Data Collection and Data Analysis

The division's top management provided *entry* to the authors for conducting fieldwork in the division, thereby bestowing immediate "legitimacy and credibility" to the researchers (Patton 1990, p. 254). An office and infrastructure support was provided during the visits to TECHCOM to conduct fieldwork. Assistance was provided by a manager and a senior staff member in setting up formal face-to-face interviews, telephone interviews, and informal meetings (e.g., lunch). Permission was also granted to attend team meetings involving distributed members,

and relevant internal documents were provided. The study was initiated in March 2006, and data collection was undertaken over a period of approximately two years (June 2006–June 2008). The formal interviews, a majority of which were taped and transcribed, were conducted in great depth (typically lasting between 30 minutes and an hour). Twenty-five formal interviews were conducted (see Table 1); in addition, the researchers had the opportunity to informally converse with some TECHCOM employees and to attend meetings. To allow the reader to appreciate the role and location of individuals quoted while maintaining their anonymity, we use the following naming convention for our interviewees: <location(s) in which the individual is/has been based>.<position><optional number to distinguish between individuals in the same location(s) and same position>. For example, the second manager we interviewed in *USCity1* was identified as *USCity1.Manager2*.

Recognizing the nascent stage of knowledge on our topic, a predominantly *inductive, interpretive* approach

Table 1 Interview Details

Respondent details	Based in	Primary role	Number of formal meetings/interviews	Other informal interactions
Head of a division (<i>USCity1.DivHead</i>)	United States	Top management	2 interviews; also 1 meeting in a group	None
Director of a division (<i>USCity1.Director</i>)	United States	Top management	2 interviews; also 1 meeting in a group	None
<i>USCity1.Managery1</i>	United States	Project management	1	None
<i>USCity1.Managery2</i>	United States	Project management	2	None
<i>USCity1.Managery3</i>	United States	Project management	2 (1 of them on telephone)	lunch meetings; other interactions
<i>USCity1.Managery4</i>	United States	Project management	2	None
<i>USCity1.Managery5</i>	United States	Project management	1	None
<i>USCity1.Managery6</i>	United States	Project management	1	None
<i>USCity1.Managery7</i>	United States	Project management	1	None
<i>USCity2.Managery1</i>	United States	Project management	None	1 (telephone)
<i>USCity2.Managery2</i>	United States	Project management	None	1 (telephone)
<i>USCity1.SysDesignery1</i>	United States	Technical architect/designer	1	None
<i>USCity1.Consultanty1</i>	United States	Development	1	None
<i>USCity1.JrDevelopery1</i>	United States	Development	1	None
<i>USCity1.SPICoachy1</i>	United States	Process improvement coach	1	None
<i>USCity1USCity2.SrStaffy1</i>	United States	Staff, coordination	1	Several interactions
<i>USCity1IndiaCity1.SrDevelopery1</i>	United States; in the past in India	Development/tech. advisor	1	None
<i>USCity1IndiaCity1.Managery1</i>	United States; in the past in India	Project management	1	None
Two other <i>USCity1</i> employees	United States	N.A.	None	1 lunch meeting
<i>IndiaCity1.Managery1</i>	India	Project management	1 (interviewed during visit to the United States)	None
<i>IndiaCity1.Managery2</i>	India	Project management	1 (by telephone)	1 lunch meeting
<i>IndiaCity1.Managery3</i>	India	Project management	1 (by telephone)	None
<i>IndiaCity1.SrDeveloper1</i>	India	Development	1 (by telephone)	None

was adopted, which has been described by Walsham (2006, p. 325) as follows:

In terms of learning from the data itself...I write impressions during the research, after each interview, for example. I generate more organized set of themes and issues after a group of interviews...I then try to think about what I have learnt so far from my field data.

Consistent with this view, and with guidelines on conducting interpretive case studies (Walsham 1995), we examined and made sense of our data, guided by the logic underlying analysis strategies associated with less procedural versions of the grounded theory methodology (Bryant and Charmaz 2007). At the core, we used the “constant comparative analysis” to identify initial concepts, and attempted to link this evolving set of concepts to higher-level categories (Charmaz 2000). We summarize the key methodological guidelines used in Table 2.

We now elaborate on the categories and subcategories, as well as tactics that emerged through the interpretation of data. As is customary in interpretive studies, we provide sample quotes to bring to life the issues being faced by the subjects themselves (e.g., Walsham 2006, Sarker et al. 2006). Over all, our study highlights that there are three broad categories of agility in the globally distributed ISD context: (1) resource-related (composed of both people and technology-based resources), (2) process-related (composed of methodology based, environmental awareness based, and temporal bridge based), and (3) linkage-related (composed of communicative relationship based, and cultural mutuality based).

4. Category I: Resource Agility

This form of agility arises from the *people* and *technological* resources at the disposal of the distributed ISD team.

4.1. People-Based Agility

People-based agility refers to the access to multi-skilled, flexible people at the different locations such that they can be deployed as and when necessary (e.g., Lui and Piccoli 2007). Our study revealed two subcomponents of this dimension of agility.

4.1.1. Ability to Rapidly Ramp Up/Down a Team.

An important aspect of people-based agility is the project team’s ability to ramp up or ramp down the size of the team (with members of specific skill sets and quality) at any given site, depending on the emerging (changing) requirements of the project. For example, to avoid project delays, it may be helpful to rapidly deploy additional team members with required skills at a given location to deal with changes in client requirements. Achieving this form of agility is inherently challenging. A senior designer (USCity1.SysDesigner1) explained:

If...we need 30 people in USCity1, it’s not going to happen...I couldn’t triple a team in 3 months in USCity1...the local headhunting agency...may have a roster of 30 or 40 people sitting on the bench...but to be honest with you, most of the ones that are pretty good are already working.

Tactics. Our data suggested that proper planning and selection of appropriate locations in which to set up overseas operations can help enhance this type of agility. For example, one way is by selecting a site (or country) that has an ample supply of skilled resources enabling the distributed team to ramp up/down a team. TECHCOM’s strategic selection of a city in India, and subsequently one in China, as key sites for shifting a significant proportion of development efforts have helped address this problem. Ramping up a team rapidly is not only made possible by employees of the company in the city/country where work may be offshored, but also by maintaining a strong pool of skilled vendor organizations at the different locations who are able to supply appropriately qualified human resources at short notice. As USCity1.Manager2 explained:

We [used] vendor team in India to do many of these systems-modifications...I was actually quite impressed in our ability to staff...team[s] up quickly and make them productive developers....

Similarly, USCity1.Manager1, who was in charge of testing, indicated that he tended to prefer vendors who had the capacity to provide adequate test resources in IndiaCity1 at short notice. Other mechanisms mentioned by USCity1.Manager2 for quick ramp-up are *rotation* (moving members between the offshore and onshore locations) and *short-term transfer* across divisions.

Table 2 Methodological Guidelines with Illustrations

Aspect of the study	Methodological considerations	Additional description	Illustration (where applicable)
Organization choice and entry	Selecting suitable organization to study intensively.	The goal was to study a “representative” organization in terms of the phenomenon, that is, ISD using distributed teams; at the same time, to study a critical case in terms of the quality/reputation of the organization, so that normative implications for future practice could be derived (Patton 1990, Flick 1998)	TECHCOM was chosen because it: (a) had significant experience in ISD across multiple globally distributed locations; and (b) is widely acknowledged as a leading organization with respect to its development and use of IS.
	Entering the field with credibility	“Known sponsor approach” resulting in immediate legitimacy and credibility of researchers (Patton 1990).	Official e-mail from senior personnel introducing the research project and the researchers to relevant division members, and setting up initial meetings/interviews.
		The researcher is not only the “observer” but also the “observed,” i.e., organizational members tend to scrutinize researchers’ actions, particularly in the initial stages (Patton 1990).	We consciously attempted to develop and maintain independent academic identity, to ensure that we were not seen as agents of management. We attempted to maintain credibility and trust by: (a) always being well prepared for the interviews, and (b) preserving anonymity and confidentiality at all costs (Myers and Newman 2007).
Data collection	Choice of interviewees	Suitable respondents were suggested by the senior staff and one of the managers who also helped set up many of the interviews.	We sought to balance both the width and the depth of perspectives by interacting with individuals in a variety of roles, drawn from different projects, different levels, and different sites/countries.
	Conduct of the interviews	“Snowballing” technique used in some instances (Patton 1990)	Relevant respondents were also identified by other respondents.
		Being sensitive to principles of: (1) “flexibility” (2) “nondirection” (3) “specificity,” and (4) “range” (Flick 1998)	(1) We happily rescheduled or shortened meetings to suit emergent demands on the interviewees’ time. (2) We tried to maintain an open and “nondirective style of conversation” during interviews. (3) We asked specific questions towards the middle of the interview and also encouraged “retrospective inspection” (Flick 1998). (4) We varied how the interviews were conducted depending on the interviewees’ motivation and ability to elaborate on agility issues.
	Researcher involvement in the study	Data collection process involved “prolonged engagement” of the researchers and their “persistent” interactions with the subjects (Flick 1998).	Our data collection was undertaken over a 2- year period, with 25 in-depth interviews, six field visits to the company, and significant informal interactions with project participants.
	Maintaining empathetic neutrality	“Nonjudgmental form of listening” (Walsham 1995, quoting Zuboff 1988); empathizing with interviewees’ frustrations but simultaneously maintaining distance (Patton 1990)	We tried to be patient and sympathetic listeners when interviewees expressed dissatisfaction with their job and boss, while not probing/seeking elaboration of the “juicy” organizational stories that were not relevant to distributed work.

Table 2 (cont'd.)

Aspect of the study	Methodological considerations	Additional description	Illustration (where applicable)
Data analysis and representation	Unearthing and refining concepts through constant comparison	In Holton's words (2007, p. 277), "the purpose of constant comparison is to see if the data support and continue to support emerging categories. At the same time, the process builds and substantiates the emerging categories..." The process is consistent with the notion of the hermeneutic circle, and involves not just "induction," but also "abduction" (Bryant and Charmaz 2007).	We used <i>in vivo</i> codes as well as theoretical concepts, and sought to refine them iteratively. Induction played the predominant role in our initial coding of data, whereas the role of abduction became more evident in the formulation of the different dimensions/subdimensions of agility and the relevant contingencies.
	Triangulation	Primarily "data triangulation" (Flick 1998, Patton 1990); comparison of responses across respondents, locations, and time as part of the constant comparison process (Charmaz 2000).	Whenever possible, we have strived to make sure that dimensions and tactics included in the manuscript have been suggested by multiple respondents in the same location, multiple respondents across different locations, and same respondent but across multiple interviews.
		Lack of agreement in triangulation not seen as indication of invalid category necessarily, but as an opportunity to include/explore differing perspectives and unearth additional contingencies (Flick 1998).	In many cases, interviews revealed disagreements among interviewees about important issues, for instance, the effects/feasibility of the "follow-the-sun" approach and also the use of agile methodologies in distributed settings. Instead of considering the disagreements as problematic, we undertook further in-depth investigation on the topics. A richer, contingent understanding emerged as a result.
	Being suspicious about evidence	Sensitivity to possible biases in interviews (Klein and Myers 1999).	Being aware that individuals in different positions and situations may bring different biases to the interview. For example, we treated with caution some overly critical views of a USCITY1 manager who had already turned in his resignation.
			We also sought to validate issues raised by such interviewees with data from other sources.
	Member checking	Validating/checking researchers' interpretations with interviewees (Flick 1998).	For example, we presented the diagram (Figure 1 in the paper) highlighting the different agility dimensions and tactics for enhancing them to a number of interviewees. We then assessed the validity of these dimensions through their reactions.
	Being sensitive to ethical concerns	1. Balancing anonymity and disclosure (Flick 1998). 2. Ensuring that the transcripts and other data were kept secure (Myers and Newman 2007). 3. Treating respondents, their knowledge, and their time with respect (Myers and Newman 2007).	1. We ensured that that the following were not disclosed: (a) organization's identity, names of US, European, or Asian cities where work was being conducted, (b) the actual technologies and methodologies that were being used, and (c) specifics about the division or any of its projects. 2. We made sure that only the authors and transcribers had access to the empirical material. 3. We readily rescheduled meetings and even shortened the interviews on multiple occasions to fit the schedules of the interviewees.

When we need to ramp up numbers quickly, I look for...“are there other teams in TECHCOM that have had to deal with this issue, and can I get some support from them?”...We brought them in, and they are allowing us to be more agile.

4.1.2. Interchangeability of Roles, Reconfigurability of the Team, and Distributed Decision Making.

Another dimension of people-based agility is the ability of the distributed team members to play different roles. In addition, this aspect of agility includes the ability to reconfigure the team and its nerve centers as and when the need arises. For example, the ability to shift the control center of the project from one distributed location to another (say *USCity1* to *IndiaCity1* or from *IndiaCity1* to *ChinaCity1*) as necessitated by constraints such as manager turnover, natural disasters, etc., can enhance the agility of a team (Grewal et al. 2007). However, in the case of B2B project teams in TECHCOM, such a reconfiguration is inherently difficult to achieve. The offshore unit members (say, in *IndiaCity1*), lacking physical and strategic proximity to the headquarters of TECHCOM (*USCity1*), do not have a clear sense of the business requirements. Furthermore, as *USCity1IndiaCity1.Manager1* pointed out, given that system requirements are necessarily formulated in *USCity1*, reconfigurability of the teams is difficult to achieve:

Our information comes from our business team. Our business is here, in *USCity1*...most of the business information is available in *USCity1* or *USCity2*. It can go only one way. So, it would be very difficult if they want to control this from India tomorrow.

In addition, a number of B2B division managers (e.g., *USCity1.Manager1*) noted that the application-related experience of offshore developers or offshore project managers, and their knowledge of the business context, is inherently much lower than personnel in the United States, even though they could very well have equivalent or even superior “technical” skills. Such imbalance in business knowledge inhibits seamless reconfiguration of the team.

Tactics. Although inherently challenging, several tactics such as encouraging team-member versatility are considered critical for enhancing this form of agility, as indicated by *USCity1.Manager2*:

I really value someone who can come in...both do web-pages and database development if they need to.

Or can jump from budgeting to server support in a pinch. I think versatility is huge...

In addition to developing multiskilled employees, having a common culture across TECHCOM’s units in different countries also allows for a smooth transition of employees when they are moved from one site to the other. An offshore employee (*IndiaCity1.Manager1*) noted:

TECHCOM, all over the world, we have the common culture...[as] if I am working in a mirror...so for people to be able to switch from one place to another...it would not be as difficult.

Although there are clearly challenges to developing this ability to rapidly reconfigure the team and to prepare offsite personnel to sense changes and respond promptly, conscious effort from the organization to disseminate knowledge across locations and foster a shared frame of reference can make it happen. Indeed, *USCity1.Manager4* noted that “in order to be agile, you really have to understand the *whole* thing.” With respect to enabling distributed decision making, one way is to invite key offshore employees onshore and help them develop the level of knowledge necessary to make independent decisions. *USCity1.Manager2* narrated his positive experience in this regard:

We had a technical...manager who came over here [from India], spent 3 weeks with us during design and went back really as my eyes and ears on the ground over there day in and day out to the point where he was able to solve many of the issues before they ever reached me, which was immensely helpful.

4.2. Technology-Based Agility

In the context of distributed ISD, technology-based agility refers to the degree of nimbleness enabled by the team members’ seamless access to comparable and compatible information and communication technology (ICT) infrastructure at each location, such that the speed of ISD is not hindered. Such ICT infrastructure may include a variety of communication technologies and groupware, centralized development platform that keeps track of the code and the bugs being detected, fault-tolerant server nodes, etc. This is sometimes difficult to achieve in cross-national distributed settings, especially because access

to development platforms and communication technologies are provided to the offshore location through telecommunication links, whose reliability depends on the country or region where the site is located. USCity1.Manager2 noted:

...their [an offshore city] networking activity was probably not as quick as ours, so transactions that take us 3 seconds to complete may take 30 seconds from the facility there.

In addition, some distributed offshore locations may have limited technological infrastructure available compared to the onshore (U.S.-based) locations. This could impact the speed and efficiency of development as well as connectivity of the site with other collaborating sites.

Tactics. Our study revealed several tactics that were successfully implemented by TECHCOM to overcome the challenges mentioned above. One way was to develop the capability to bypass the local public infrastructure, when necessary and feasible. Another way is to invest in a globally standardized technological infrastructure that enables locational transparency, as USCity1.Manager2 explained:

We have wireless networks in all our facilities...you could walk in...to the [EuropeCity1] and [IndiaCity1] offices.... I walk in and my laptop works there...I think a lot of other companies struggle with that because their "car keys don't open the doors in the right places"...and that...drags the whole project down.

Another employee (USCity1.SPICoach1) echoed this point:

You can be working in [any location]. Your access to information is not location dependent and it has been this way for almost...five years...that has given us tremendous flexibility.

One of the key characteristics of distributed teams is their reliance on communication technologies for work accomplishment, which can be challenging. USCity1.Manager1, for example, stated "the communications, the infrastructure, the phones [offshore] aren't easy to hear clearly..." Thus, another way to improve technology-based agility is to make available a portfolio of communication technologies that works for the entire distributed team in different circumstances. It needs to be noted that perceived "richness" offered by a communication technology depends not

only on the objective properties of the technology media, but also on how it is being appropriated (Lee 1994). Several interviewees reported that relatively less "rich" technologies (i.e., a groupware with shared desktop feature coupled with audio) were more helpful in understanding their distributed colleagues compared to videoconferencing. USCity1.Manager1 stated:

[Tool Name]...that's been helpful where you know for sure that you are looking at the same page of the word document, or...PowerPoint presentation page, as the person who's presenting.

Finally, given that the core focus of a distributed ISD team is on the technical development of a software application for the client, a key tactic for increasing technology-based agility for such teams is to provide them with a centralized development platform that keeps track of the code and the bugs being detected and provides seamless access to this information at all locations. USCity1.Manager3 provided details of such a technology used in TECHCOM that helped in the development.

[The tool] tracks all the bugs for a particular project... Also, it does all the code management, the source code management. It all integrates into one big [repository]...it's a great tool...because every single team member can log into a bug, can update it, can add work content or details on the bug and things like that.

He indicated that distributed teams in some organizations "track all their bugs in Excel" and "were sending them by email" significantly lowering the speed of response to bugs detected. In addition, IndiaCity1.Manager2 pointed to technologies and services provided by vendors that ensured backup and speedy recovery of work in the globally distributed environment.

5. Category II: Process Agility

This pertains to the agility that originates in the team processes, such as: (a) the systems development methodological practices guiding the project; (b) environmental scanning and sense-making routines for anticipating and/or recognizing possible or imminent "crises"; and (c) the organization and transition of work across time zones.

5.1. Methodology-Based Agility

Such agility arises in two primary ways, which we discuss below.

5.1.1. Adaptation of Traditional ISD Methodology/Practices. In its distributed projects, B2B division has historically used TECHCOM's proprietary version of the SDLC that is blended with prototyping (SOFT-METHOD, a pseudonym). The use of the time-tested "industrial strength" SOFT-METHOD often involves (roughly) the following four steps in the distributed environment: (a) application managers interacting with the client and preparing detailed specifications and the design; (b) prototyping by developers in one of two U.S. locations, and feedback from the customers and application manager, leading to the refinement of the specifications and the design; (c) undertaking the development offshore with application-related decision making from the United States; and (d) testing by onshore/offshore professionals, coordinated by testing managers who were usually located in the United States. Although the process works well overall, many TECHCOM professionals pointed out the lack of agility of the process. For example, the USCity1.SysDesigner1 questioned the value of too much analysis and design upfront before sending the work offshore:

Agile people say there's a four letter word called BDUF...it means Big Design Up Front...and we call it a four letter word because literally you don't do Big Design Up Front...another acronym [is] YAGNI, You Aint Gonna Need It.

Tactics. For enhanced effectiveness as well as the agility to react to changing customer requirements, the use of prototyping had to be expanded, not only involving team members in the United States but also involving the participation of offshore developers, as noted by USCity1.SysDesigner1:

We actually have prototype developers who work in USCity1 and some in USCity2...they will work directly with the customers.... You then throw [the requirements] over the wall at IndiaCity, well...ok, you've lowered the amount of guesswork that the resources in IndiaCity have to use, but then there's still a 3–4 month window that they are writing code, so during [those] 4 months, [the assumption is that] nothing changes. Well, the business changes...in IndiaCity, they are busy...going into the covers and writing the code [without realizing that changes have occurred].

Furthermore, USCity1.SysDesigner1 suggested that agility may be enhanced by training offshore developers and technical project managers to think in terms of "business value" of the work they do, rather than in terms of "I am a techie...I want to make that JavaScript work real well." Yet another aspect of the methodology highlighted by him is the need for consistent use of diagramming techniques across the locations, rather than relying solely on text, which can be interpreted in multiple ways. Equivocal text can make it difficult for distributed members to rapidly converge on a meaning, especially when changes have to be introduced and communicated quickly across locations (Sarker and Sahay 2004). IndiaCity1.Manager3 offered yet another tactic for the use of the traditional methods in distributed ISD contexts. According to him, using a combination of traditional methodologies with some of the basic practices from agile methodologies has worked successfully in some distributed ISD projects in TECHCOM. Specifically, IndiaCity1.Manager3 highlighted that his team followed a waterfall model, but froze the requirements in four-week time periods (instead of longer durations, which is more characteristic of the traditional waterfall). Within this period they engaged in "daily stand-up meetings," often with the client present. Such shorter time periods enabled the developers, managers, and the client to have a better view of the project status and be more agile to changes, compared to teams using traditional waterfall models, where clients do not get to see a working product for a very long period of time.

5.1.2. Appropriately Harnessing Agile Methods. Agile methods have been found to reduce the challenges associated with temporal, geographical, and sociocultural distances (Holmstrom et al. 2006). Thus, not surprisingly, some teams at TECHCOM have also tried agile methods such as Scrum. IndiaCity1.Manager2 stated:

All...software development is moving from waterfall...towards agile. Customers are asking for SCRUM...in the very early stages of the project [customers] can see something coming up whereas in the waterfall, they have to wait for 6 months before they see something coming up, and by that time the requirements have already changed, because the business has changed.

Interestingly, interviews indicated that the use of agile methods has not been very effective in some cases. One reason, according to the USCity1.SysDesigner1, who was considered by many B2B colleagues as an agile methods expert, was the high degree of customization of agile methods (e.g., XP or Scrum) attempted in some of the projects he/she had been involved in:

I don't think you are going to benefit by taking XP which is 12 rather rigid practices and pulling 6 of them and saying these are the 6 we'll do...if you take enough out of Scrum...it turns into waterfall. If you take enough out of XP it turns into waterfall too or spiral.

USCity1IndiaCity1.Manager1 noted that the adoption of agile methods magnified the problems associated with distributed work, seriously denting the work-life balance of key employees:

We want everything faster than originally scheduled. There is a lot less focus for documentation. People even describe what is required over the phone, and someone there captures in their notebook and someone starts building it the next day. Again, for the communication to happen, it will mean people staying late there or our people are coming early here. People call from home...Usually when you go home, you have other responsibilities to take care of...

Again, Scrum seemed to be burdening team members with more demands on time and constant pressure; however, team-members at either location could see no apparent benefits accruing because of the team's adoption of Scrum. Not surprisingly, in the words of USCity1.SysDesigner1, "the developers abandoned Scrum just as quickly as they could get off of it," and according to USCity1.Manager6, some organizational members had a "pre-conceived notion that Scrum was evil," and even considered it "Scum." On a more rational note, another manager (USCity1.Manager7) explained why he was reluctant to implement Scrum in his distributed project teams in the first place:

We have nothing [historical performance data] to base what a Scrum model could deliver...and I don't want to be the first one to sign up for that process.

Yet another issue the USCity1.SysDesigner1 highlighted had to do with the lack of training on how to use Scrum, which led to its abandonment in a project involving U.S.-based and Indian staff:

They [the Indian team-members] were not familiar with Scrum. So, when we asked for "numbers"

[everyday], they didn't know what numbers to give us.... [They asked] "Is all I am doing re-estimating everyday? That's just dumb." So the numbers would only get updated once or twice a week, and usually badly. So the "burn-down charts" were just useless.... Why did they abandon it? Because they weren't trained to use it....

Tactics. In light of the above, USCity1.SysDesigner1 indicated that there was a need for a high-profile Scrum evangelist and trainer at both locations (especially offshore) who would explain the approach, tangibly illustrating the initial challenges and benefits, cultivate a positive image; ensure that agile methodologies are not modified to the extent that they reduce to traditional methodologies; and train teammembers and get them excited about using it. Furthermore, there was a need to be sensitive to the additional complications caused by the time-space distances including work-life balance issues. Finally, collecting and providing performance data associated with agile methods use in distributed settings could be beneficial in encouraging their adoption.

5.2. Environmental Awareness-Based Agility

This pertains to the systemic ability in the team to anticipate, recognize, and react to changes in the project because of turnover, organizational changes, etc. Largely, this capability is a function of the sophistication of in-built processes for scanning the client and partner organizations, sense making, rapid transmission of scanned and processed information, and decision making (e.g., Grewal et al. 2007) within the distributed team. For example, a common concern facing organizations such as TECHCOM that rely primarily on offshore insourcing but have not expanded their own staff at offshore locations sufficiently to be able to handle *all* the work assigned, is that there may be changes in the ability of vendors to provide adequately qualified staff, because of tight human resource market conditions and turnover.

Tactics. This element of agility can be enhanced by monitoring the possibility of vendor staff working for TECHCOM's projects being unavailable in the near future, as USCity1IndiaCity1.Manager1 recounted:

We used to work with 4 or 5 different [vendors] in India. We had a manager from each company come in and tell us at a very high level...in a green or red kind

of status in terms of the continuity [of their employees] once in 2 weeks.

IndiaCity1.Manager1 noted that by maintaining good relationships and constant touch with the vendors, TECHCOM usually was able to pick up necessary signals to prepare for changes:

...relationship plays an important role there, so the idea being that our account manager and their relationship manager should converse...in case of turnover and in cases like big company changes, there is...communication [in this regard].

More broadly, IndiaCity1.Manager2 emphasized the need to closely monitor the project environment by arranging weekly meetings with representatives of all locations, all vendors, and key stakeholders in the client organization. This would enable advance signal detection and response design.

5.3. Temporal Bridge-Based Agility

This aspect of agility refers to the ability of the team to collaborate across locations without significant temporal delays, and, when possible, by taking advantage of time zones.

5.3.1. Seamless Transition of Work Sequentially Across Time Zones. The attractiveness of this model of work is that nighttime, which is usually lost, can be reclaimed as productive time by team members at the other end of the world (Sarker and Sahay 2004), thereby speeding up relevant ISD-related processes, and indirectly contributing to the team's agility. Indeed, one of the managers (USCity1.Manager5) noted that follow-the-sun "...works. We have been working in this way for quite some time." IndiaCity1.Manager2 stated that:

everybody tries to take advantage of the global delivery model...24 hours of the clock...though [in] hardcore development...you don't see that too much.

Tactics. USCity1.SysDesigner1 argued that to ensure seamless transition of work it is important to use the right kind of projects and appropriate allocation of tasks for which such a transition works. He added:

I think you can do it only for the simplest tasks, where the amount of communication that has to transition across time zones when you are going from India, to Europe to the USA...is so thin that you can accomplish it...

USCity1.IndiaCity1.Manager1 provided a more cynical view on the 24-hour development myth, emphasizing the need to be skeptical of vendors' "follow-the-sun" gimmicks:

...every company [i.e., vendors from a certain country] says they follow the model...companies have their employees working in the US during the night, and sending the information back to them [clients]...showing that it is coming from [offshore].... They [the offshore vendors] say that... they follow the sun and it works like that...after they build the trust, then the client thinks yes, this works...then they finally move it back...[offshore].

Other respondents added that to enhance this form of agility, it is imperative that the team members across the locations have uniform work-practices and similar norms of interaction for developing a shared understanding of issues. Finally, the need to locate the right personnel/roles in time zones consistent with the flow of work was highlighted as an important tactic for enhancing this form of agility by USCity1.Manager3, who had his developers in China and testers in India:

I think we have some of our teams that are just in the wrong model [place]. It would probably be beneficial to have...my test team in China, do the testing so that they can identify the bugs and India, a few hours later, could actually resolve those bugs.... Unfortunately...we are doing stuff almost a day later.

5.3.2. Capability to Minimize Challenges Because of Time Differences Among Locations Through Innovative Work Practices. One challenge facing distributed teams is time differences across locations, which can cause delays in receiving responses and clarifications, and therefore delay the flow of work. USCity1.SysDesigner1, who had obviously been frustrated by the delays in receiving clarification from colleagues located in far-off time zones, wondered:

Who do you ask? If...there is not someone local you can ask and it's 9 A.M. in the morning, then you'll send out your e-mail, and you won't get your response until at earliest, the first part of the next morning.

Tactics. Although time zones cannot be changed to reduce time delays or responsiveness, what distributed teams can do is to engage in frequent synchronous meetings, in an effort to enable rapid and immediate feedback, and avoid misunderstandings.

However, it is essential that in scheduling these meetings, team managers are even-handed and considerate to physiological and social clocks of offshore members, as IndiaCity1.Manager3 stated: “[select a time] where all teams can meet.”

In very urgent situations, the notion of “swap team” is invoked. USCity1.DivHead explained how the challenges posed by temporal delays are addressed by shifting work times:

...they'll have developers in India and developers in US-City 1 that can address defects at the different times of the day so we don't end up with a 2 or 3 day delay to get a defect fixed.... The other things that we've done, is...towards the end of the cycle, we may ask the US-City 1 team or the India team for 4 or 5 days switch the time that they are working so the India may come in on India time for the second shift which would be for USCity1 during the day.

Naturally, as IndiaCity1.Manager2 noted, such arrangements are not sustainable over the long run, because they lead to serious work-life balance concerns, and raise employee turnover possibilities.

6. Category III: Linkage Agility

This pertains to the agility that originates from the nature of interaction-based relationships that a team enjoys within itself or with key external constituents. Two components of linkage agility emerged in our study: cultural mutuality-based agility, and communicative relationship-based agility, which we discuss below.

6.1. Cultural Mutuality-Based Agility

This aspect of linkage agility pertains to the extent to which the cultures of the team members seamlessly mesh together such that there is minimal hindrance to the development of shared understanding and mutual relationships (e.g., Holmstrom et al. 2006, Sarker and Sahay 2003). Our study revealed two facets of this agility.

6.1.1. Intercultural Competence of Members in All Locations. Historically, culture has been associated with the physical “place” of individuals. Again, owing to the geographic dispersion (and associated cultural dissimilarities) amongst members in distributed teams, intercultural awareness does not occur naturally. TECHCOM faced several

challenges in developing this form of agility. As USCity1USCity2.SrStaff1 noted:

I think especially the India relationship is even more difficult [than a country in Europe] because.... There's a language barrier. There definitely is. I am sure our accents to them are just as difficult to understand as theirs are to us. Um...there are cultural differences...different work ethics.

Along similar lines, USCity1.Consultant1 observed:

I see my colleagues struggle...they talk with someone in India, for example, and they [the Indians] want a lot of details. They [Indians] take apart spec 1 page by page, line by line. For some of my colleagues, that is frustrating because they are like, this is obvious, how come we are not getting...[through]?

Another cultural issue that seemed to have had an impact on the team's timeliness and speed was described by USCity1.Manager1:

I know in certain cultures...to say “no” in a certain situation is culturally basically unaccepted so they're going to say “yes,” even though they may not be able to do this....and, so you think “oh good, that's not a problem.”.... Suddenly when the date hits, it's not ready...not because they...haven't been working...hard, but because it was unrealistic.

Such situations can have a negative impact on relationship/linkages and could cause unexpected delays in responding to changes. Although processes of globalization may be leading to a degree of homogeneity of cultural practices and meanings, there are still cultural differences amongst distributed members. Indeed, India City1.Manager2 noted that having cultural mutuality “absolutely...moves the project faster.”

Tactics. One way to mitigate teams' sluggishness due to cultural problems is for members to have an open mind and to participate in distributed work with the awareness that one will be dealing with individuals from different cultures, as suggested by USCity1.Manager1:

...when you're dealing with different cultures, if you go into it thinking that everyone is going into it thinking like we do in...the United States, that is a big mistake. I think you need to recognize...what we may take for granted might be entirely different than what they may take for granted.

IndiaCity1.Managers2 and IndiaCity1.Managers3 both also indicated that the *cultural competency training* they attended at TECHCOM was very helpful in

developing a productive linkage with distributed team members. Yet another tactic for increasing the intercultural competence of the members is to temporarily relocate selected members from IndiaCity1 to USCity1 and vice versa during a project, as was suggested by USCity1.Manager7. Furthermore, IndiaCity1.Manager3 suggested that a genuine effort to get to know distributed members often resulted in much faster responsiveness from them.

6.1.2. Common and Professional Work Culture Within the Entire Organization Across Geographic Locations. The other element of cultural mutuality-based agility is to have a common work culture within the entire organization, irrespective of the location.

Tactics. As IndiaCity1.Manager3 suggested, the primary tactic through which to develop a common culture is to ensure that “everybody goes through a [similar] training.” Another tactic would be to ensure that individuals who are hired adopt a professional approach in their interactions with distributed members from other cultures, being conscious of any cultural biases they may have, and actively seeking to mitigate the effects of these biases. USCity1.Manager3 explained:

...the majority of the people that I have interacted with in TECHCOM...are professionals...you don't talk about politics, you don't talk about religion at work. I tried to make sure that everyone on the team knows that I am open and I can talk about anything. I am open, just because I am American doesn't mean I voted for [a particular politician]. Things like that. But for the most part it is very professional and we don't let those things get in the way...

6.2. Communicative Relationship-Based Agility

Communication linkage-based agility among distributed team members refers to the ability to seamlessly communicate with each other. In a distributed team context, where members communicate primarily through the use of computer-mediated communication media, this form of agility is enabled in three different ways, which we discuss below.

6.2.1. Continuous Awareness (and Visibility) of Distributed Team Members. One of the keys to relationship development leading to communication agility is to be able to feel the continuous presence

of team members. In a distributed ISD setting, there is a lack of physical situatedness among team members, which hinders effective communication, coordination, and development of cohesion. The inability to physically verify the actions of remote members seems to breed a sense of uncertainty and suspicion in some cases, which negatively affects the speed with which a team can react to a directive (Sarker and Sahay 2004). One of our interviewees, USCity1.SysDesigner1, noted:

It was very frustrating because I would hear one thing but not see results that were in accordance with what I was hearing. I was hearing what made sense, and what I wanted to hear, but I saw something different in terms of the results... Techies like to hide in corners... the vast majority of techies who hide don't deliver good stuff. How do you reach across the phone and strangle somebody... there have been times when I wanted to do that!

Tactics. IndiaCity1.Manager1, based in India, however, seemed more optimistic, and noted specific tactics that may help alleviate such problems. One of them could be by having individuals at the two locations whose primary responsibility is to keep each other updated with the daily activities at each site, in an effort to reduce the challenges associated with a lack of physical situatedness:

The [distributed] model works best when the key people... the people who are maintaining that relationship between on site and offshore, work hand in hand. They are like shadows of each other... agility is taken care of then.

Furthermore, in his/her view, by sharing electronic “space,” even though not the same geographical “place,” an adequate degree of shared understanding and visibility can potentially be maintained:

The development team is located in India, so even if they are there [in India] right now, they are on-board... We have shared [relevant information about the project] on the site which is accessible by everyone. Whatever documents are prepared, we upload them there... so anyone who has access, wherever in the world they are, they can still know... what design we are doing... despite the geographical distance, it's one team... working on the project.

6.2.2. Maturity of the Interlocational Communicative Relationships. Another component of communicative relationship-based agility is to have a highly matured communicative relationship in the distributed team members (Sarker and Sahay 2003). With respect to his team spanning United States-India-China, USCity1.Manager6 observed:

...I would have to say we didn't do so well with agility because the team wasn't formed well. It was still immature in terms of its ability to communicate and interact with one another. As the team became more mature, it was easier...to adapt to changes in requirements...

Tactics. Such a maturity in the relationship can be developed by building a social connection, encouraging the gradual development in trust amongst the team members through team-development exercises and social activities. USCity1.Manager3 provided an example of such a tactic that was successful:

I think you could talk to anyone that went on the trip when we went to India. We might not have gotten a lot of work done, but it was the dinner that we had with our co-workers that night that was more beneficial than actually any meeting that we had where we were actually working on the project.

6.2.3. Close Collaboration Among Clients and Distributed Team Members. Communication link-age-based agility should exist not only amongst the team members, but also between the team and its customers. It is important to understand the customer requirements, provide the customer access to the team when needed, and react rapidly when requirements do change.

Tactics. Conscious efforts to develop close relationships with customers and other key stakeholders related to the systems development project can help address such challenges. Specifically, seamless connectivity to the customers can also be developed by having open lines of communication and by assigning selected members in close physical proximity to the customers. IndiaCity1.Manager1 elaborated:

...in a distributed model, irrespective of where work is done...the client should be able to call up and say..."I had this idea today. Are you doing it this way or not? This is how I thought it was going to be documented, but this is not how I wanted." So, it's important...because the team is sitting off-shore...someone from the team should be in close proximity of the customer.

On similar lines, USCity1.SPICoach1 added that customers need to be motivated to stay in touch:

We have to explain to the customer that if you are looking for the benefits of agility...you have to make the sacrifice to be available at an odd time of the day.

We summarize the above discussion in Figure 1.

7. Exploring Interrelationships and Contingencies Among Agility Facets

Note that Figure 1 does not provide any indication on the contingencies as to when a particular dimension could become more or less salient. We believe that such guidance would be potentially valuable to practitioners deciding on which agility to invest in, and thus we offer some preliminary findings in this regard. As a first step, we sought to understand those key contingencies surrounding a distributed ISD project, for which the dimensions of agility would have varying degrees of salience. Perspectives of our interviewees on this issue varied. For example, IndiaCity1.Manager2 argued that if you have really skilled individuals in a team, the agility dimensions identified would be important, but not critical, to the success of the project:

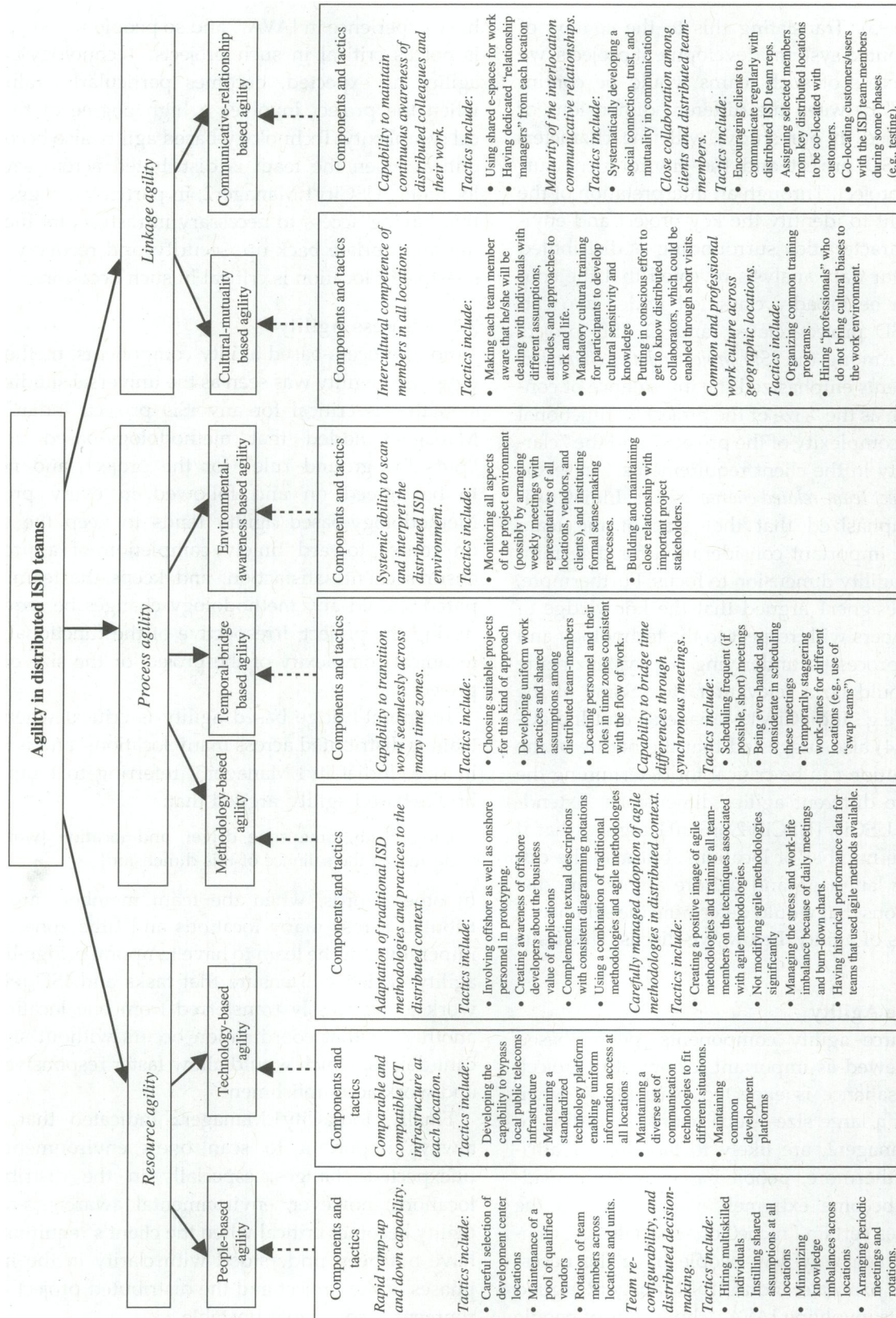
you put several Michael Jordans...in one team, and everything will happen [get done irrespective of the degree of agility].

However, having a team full of "Michael Jordans" (i.e., superstars) is not easy to build, and rarely happens in practice; thus, agility concerns need to be carefully addressed for distributed ISD teams. In sharp contrast to the views of IndiaCity1.Manager2, IndiaCity1.Manager3 argued that:

Everything is important. If you remove any of these parameters [i.e., the agility dimensions], you are risking your project.

As mentioned earlier, developing all aspects of agility may not be feasible (or even desirable) for every distributed ISD team, and it is thus useful to identify the specific contingencies under which certain aspects gain/lose relevance. To contribute toward this objective, we turned to the contingency theory perspective, which argues that a fit between the *organization's elements* with that of the *organization's environmental characteristics* would determine its success

Figure 1 Elements of Distributed ISD Teams Agility and Selected



(Donaldson 1995). Translating this to the context of globally distributed systems development projects, we argue that success of such teams would be contingent on a “fit” between the different dimensions of agility (identified in our study) with the characteristics of the ISD project and the environment surrounding the project. Through an interpretation of the data, we sought to identify the key project and environmental characteristics surrounding a distributed ISD project. Our data analysis revealed that the contingencies can be broadly classified under two categories: (1) ISD project-related, and (2) ISD project team-related. Among the *ISD project-related characteristics*, respondents emphasized the importance of contingencies such as the “size of the project,” “functional and technical complexity of the project,” and the “clarity and stability in the client requirements.” In terms of the *ISD project team-related characteristics*, IndiaCity1.Manager3 emphasized that the “size of the team itself” was an important consideration for determining on which agility dimension to focus. Furthermore, USCity1.SysDesigner1 argued that the knowledge of the team members with respect to the technology and the business processes underlying the system being developed would be a relevant factor. Some of the interviewees (e.g., IndiaCity1.Managers3 and IndiaCity1.Managers4) also suggested that the nature of the team’s distribution can be critical for determining the salience of the different agility dimensions. Extending this idea, USCity1USCity2.SrStaff1 argued that it is not only the number of locations, but also the difference in the language and culture among the distributed locations that would determine the salience of certain aspects of agility. Below, we discuss this issue in some detail.

7.1. Resource Agility

Among resource agility components, people-based agility was viewed as important for any ISD project; however, its salience is expected particularly when the team is of a large size—such teams, according to IndiaCity1.Manager2, are likely to face higher attrition rate. Furthermore, people-based agility considerations can become extremely valuable when the project involves certain “niche” or “specialized” technology, for which very few people have the necessary skills. For example, IndiaCity1.Manager2 argued that for projects involving JAVA, “thousands of people

have experience in JAVA,” and so people-based agility is not as critical in such projects. Technology-based agility, as expected, becomes particularly valuable when the project involves a high degree of technical complexity. Technology-based agility also becomes critical when the team is distributed across several locations. USCity1.Manager2, in particular, suggested that having access to necessary infrastructure including appropriate back-up, security, and recovery irrespective of location is critical in such occasions.

7.2. Process Agility

Among process-based agility components, methodology-based agility was seen as the universal subdimension that is critical for any ISD project. IndiaCity1.Manager3 added that methodology-based agility “puts the ground rule” for the project, and needs to be agreed on and followed in every project. Methodology-based agility tends to keep the team on course toward timely completion of a project, ensures client satisfaction, and keeps the team prepared should any methodology changes be required during the project, irrespective of the functional and technical complexity of the project or the size of the project.

Temporal-bridge based agility is critical when the team is distributed across many locations, and is large in size. IndiaCity1.Manager3, referring to temporal-bridge based agility, argued that:

Large team, time zone driver, and location [would decide on the salience of this dimension].

In other words, when the team members are distributed across many locations and time zones, it is imperative for the team to have temporal bridge-based agility, which will ensure that tasks and ISD-related work is seamlessly transferred from one location to another, or that coordination occurs without significant delays, which would allow faster responsiveness and work accomplishment.

Finally, IndiaCity1.Manager2 indicated that it is always important to scan one’s environment for unexpected changes, especially in the distributed locations; however, environmental awareness-based agility becomes critical when the client’s requirements have not been understood with clarity in the initial phases of the project and the distributed project environment is somewhat unstable.

7.3. Linkage Agility

The importance of communicative relationship-based agility was acknowledged to be critical for any globally distributed ISD project; however, it becomes even more critical when the team is distributed across locations where the commonality of language (say, English) is not very high, resulting in severe challenges in everyday communication. USCity1.Manager2 stated:

I'd say with teams that are located in native-English speaking countries like Ireland, we rarely have any difficulty...but with the Brazilians and especially the Chinese vendors now, we face a significant level of communication difficulty.

Communicative relationship-based agility was also seen as critical when there was a lack of clarity (or instability) in the client's requirements.

IndiaCity1.Manager1 added:

one of the prerequisites...is...clarity of the requirements...it's not what you have on paper which matters...so that personal relationship with the customer is always very important...

In addition, this type of agility becomes relevant when the knowledge levels across locations, whether application related or technology related, are significantly different, as in USCity1 and IndiaCity1. Finally, cultural mutuality based agility is critical when the team members are situated in *vastly different cultures*. IndiaCity1.Manager3 contended that in such cross-cultural situations, cultural mutuality-based agility would be "super important."

In Table 3, we present the definitions of the different dimensions/subdimensions of agility for easy reference, and also list some of the contingency

Table 3 Dimensions/Subdimensions of Agility and Sample Contingencies

Dimension of agility	Definition/description	Sample factors contributing to importance of subdimensions
<i>* Subdimension</i>		
Resource agility	Agility arising from the team's access to appropriate resources.	
<i>* People-based</i>	Agility arising from the access to "multiskilled, flexible people" at the different locations for rapid deployment as and when necessary.	Large size of team necessary (at times); high attrition of key personnel; specialized skills required.
<i>* Technology-based</i>	Agility enabled by the team members' seamless access to comparable, compatible, and relevant information and communication technology (ICT) infrastructure at each location.	Technical complexity; backup and security critical; several locations.
Process agility	Agility arising from the team's project-related processes/practices.	
<i>* Methodology-based</i>	Agility arising from the appropriate adaptation of traditional ISD methodology/practices, and proper harnessing of agile methods.	Universal
<i>* Environmental-awareness based</i>	Agility arising from the systemic ability in the team to anticipate, recognize, and react to changes in the project.	Turbulent/unstable project environment; requirements unclear.
<i>* Temporal bridge-based</i>	Agility arising from the ability of the team to collaborate across locations without significant temporal delays, and when possible, by taking advantage of time zones.	Many time zones; large size of team; far-flung locations; time pressure.
Linkage agility	Agility arising from the nature of interactional relationships within the distributed team and between the team and its customers.	
<i>* Cultural mutuality-based</i>	Agility arising from the team members being part of a seamless mesh of cultures such that there is minimal hindrance to the development of shared understanding and mutual relationships.	Large cultural distances/differences across locations.
<i>* Communicative relationship-based</i>	Agility arising from the ability to seamlessly communicate with each other, including continuous awareness of the actions/reactions of distributed team members, mature communicative relationships across locations, and seamless connectivity with the customers.	Number of locations; different language and knowledge levels across locations; no clarity or stability in requirements.

factors that appear to enhance the importance of each subdimension.

8. Discussion

8.1. Contributions

In this study, we have interpreted the experiences associated with the distributed ISD model of a division in TECHCOM, identifying the core elements of globally distributed ISD team agility, and how they are harnessed. Below, we highlight the specific contributions of this work.

8.1.1. Developing a Conceptual Framework Surrounding the Different Dimensions of Agility of Distributed ISD Teams. As mentioned in the introduction, there is a limited understanding of agility and its components in a distributed ISD setting, and indeed past literature highlights that the concept of agility may be paradoxical for a distributed setting (e.g., Simons 2006). Our intensive study attempts to bring some clarity on the topic, revealing that multiple dimensions of agility exist, many of which are uniquely applicable to the globally distributed ISD setting. Drawing on the past literature and our in-depth empirical study, we now provide a more comprehensive understanding of agility. Overall, the study indicates that agility in a distributed ISD setting *is the capability of a distributed team to speedily accomplish ISD tasks and to adapt and reconfigure itself to changing conditions in a rapid manner* by (a) drawing on appropriate IS personnel and technological resources; (b) utilizing appropriate ISD methodologies, mechanisms for bridging temporal distances, and routines to anticipate, sense, and react to changes in the distributed team's project environment; and (c) forging and maintaining linkages across communicative and cultural barriers existing among the distributed team members. The study also offers actionable tactics that may be implemented to enhance agility in distributed ISD teams. We are hopeful that the granularity of ideas in the understanding of agility will enable: (1) a pointed assessment of a given team's agility, and in the future, the possible development of a standard instrument to assess distributed teams' agility; and (2) the application of some of the tactics discussed in the manuscript to enhance agility. We also hope that the results of the study will encourage researchers to focus further

efforts on identifying, refining, and validating tactics for increasing agility in distributed teams.

In identifying and clarifying the elements of agility, and presenting an inductively developed conceptual framework for agility in the distributed ISD setting, we believe that the study makes an important contribution to the existing literature in the form of a "Type I" theory (Gregor 2006). Such theories go beyond mere descriptions, by "analyzing or summarizing salient attributes of phenomena," and are particularly valuable "when little is known about some phenomena" (Gregor 2006, p. 623), as in our case. Although our study does not claim to offer a definitive theory of agility in a distributed ISD context in a positivistic sense with a coherent system of causal relationships (Sutton and Staw 1995), we believe that our "interim struggle" (Weick 1995) has uncovered "tendencies" (Walsham 1995) in terms of the types of relevant agility dimensions, likely tactics, and impacts on speed and adaptability of distributed ISD teams, thereby contributing to the valuable process of initial theorizing on this topic (Weick 1995).

8.1.2. Providing a Contingency Perspective Surrounding the Role of the Agility Dimensions in Distributed ISD. We believe that our study actually goes beyond a "Type I" theory by recognizing the practical challenges of simultaneously developing/nurturing/maintaining different identified aspects of agility in distributed ISD projects, and *attempting to uncover provisional contingencies under which a particular aspect of agility would become more (or less) salient*. Our effort is consistent with that of Harris et al. (2009), who identify boundary conditions for their model. Specifically, the study highlights that factors such as the complexity of the ISD project, clarity and stability of the client's requirements, and the nature of the distribution of the team can determine the relevance and importance of the different agility dimensions. We believe that this is an important contribution, because it can potentially guide practitioners about aspects of agility that they need to focus on in a particular project. However, we must characterize the contingency findings as preliminary, and future research needs to be undertaken to refine knowledge in this area.

8.1.3. Clarifying That Agile Methods Use Does Not Equate to the Existence of Agility in Distributed ISD Teams. By recognizing agility in distributed ISD

teams as multifaceted, and identifying its contributing elements (Figure 1), this work highlights how achieving agility of distributed ISD teams is distinct from (and more than) merely utilizing agile methods, such as Scrum or XP. Indeed, our study reveals that *the application of agile methods is just one of the potential sources of achieving methodology-based agility*. Moreover, there is an expanding body of research that points to the difficulties in utilizing agile methods in distributed teams. Consistent with this view, during our fieldwork, we found that many of the guidelines or prescriptions for agile methods could not be used effectively (and sustained) in distributed ISD teams at TECHCOM. For example, one of the key components of Scrum is daily Scrum meetings (e.g., Murphy 2004). Such daily meetings are difficult to hold or organize in distributed settings where stakeholders are (often) located in different time zones. Another characteristic of Scrum is the use of Scrum sprints (i.e., achieving short deadlines), and burn-down charts that track, on a daily basis, the number of hours left on the different activities initially targeted for completion in a particular sprint (Murphy 2004). A number of our interviewees suggested that maintenance of such daily burn-down charts and stringent monitoring rules can become difficult in a distributed setting, because of a variety of unanticipated complexities associated with local cultures, infrastructures of varying quality, and temporal differences; in some attempts at using agile methods at TECHCOM, this led to dysfunctional stress and work-life balance challenges, with negative impact on morale and productivity.

Overall, our study suggests that adoption of agile methods is *one of the several potential contributors* to distributed ISD team agility, and that agile methods *need suitable adaptation* for effective adoption/use in a distributed ISD setting (as shown in our earlier analysis).

8.1.4. Highlighting That the Implications of Our Analysis of Agility Issues Go Beyond Current Guidelines for Effective ISD. A broader contribution arising from the analysis of distributed teams from an agility perspective is that it provides a framework and language for improving distributed ISD team processes. The improvement of ISD processes and the quality of software has been a core concern of the

information systems community, owing to the “long-standing reputation for poor quality in their [information systems] development and delivery” (Smith 2004). The Software Engineering Institute (SEI) was founded to address this issue, and its efforts led to the development of the Capability Maturity Model (CMM), which “is a conceptual framework that represents best practices in software process management” (Smith 2004). The CMM proposes “five stages” or “maturity levels,” with each stage providing “rigid” guidelines as to the specific activities to be undertaken to ensure that the systems development process is efficient, effective, and leads to a high-quality software product. The popularity of CMM notwithstanding, prominent pioneers in this area have criticized the program, saying that CMM has a *checklist* character—it tells organizations *what* to do, and not *how* to do it (e.g., Humphrey 1999). In an effort to address some of the deficiencies of CMM, Humphrey spearheaded efforts toward developing the Personal Software Process (PSP), where the goal of PSP was to help IS developers enhance the efficiency/effectiveness of individual-level systems development practices (Humphrey 1999). Although this was helpful, subsequent realization that ISD is rarely undertaken by a single information systems developer, but mostly by teams of IS developers, led to a reexamination of the tenets of the CMM and PSP. The Team Software Process (or TSP) came to be formulated, and widely adopted in organizations, including TECHCOM. TSP (with PSP) provides guidelines for developing “self-directed teams,” for contributing as effective team members, and for estimating time for different ISD activities (i.e., design, development, testing) in the teams.

Undoubtedly, CMM, PSP, TSP, etc. have been invaluable in informing the management of ISD, with PSP and TSP in particular emphasizing the *people* aspects. However, CMM, PSP, and TSP guidelines assume a stable environment and focus on upfront planning, and thus offer little guidance when it comes to frequently fluctuating/changing requirements and project contexts (see Table 4). Furthermore, they provide limited (if any) direction for enhancing the nimbleness and speed of distributed teams involved in ISD, which face unique challenges arising out of time-space distances amongst the various stakeholders (e.g., Sarker and Sahay 2004, Carmel 1999). In fact,

Table 4 Guidelines Offered for Effective Distributed IS Development Given Specific Contingencies

ISD process improvement practices → Key contingencies affecting globally distributed ISD teams		CMM	PSP	TSP	This study
ISD project-related	Large project	X	X	X	X
	High functional and technical complexity	X	X	X	X
	Low stability and clarity in client requirements				X
ISD project team-related	Team members distributed across two or more locations				X
	Large differences in the language and culture (both national and business related) among team-members				X
	Large size of the project team			X	X

some practices of PSP and TSP were implemented in TECHCOM, but according to our interviewees, neither were the guidelines adapted for the distributed context, nor was there much formal understanding on how to collaborate in a distributed setting. In fact, one of the respondents conceded that he was pretty much “making...up” practices for enhancing agility and effectiveness of his distributed ISD team. Along similar lines, another participant emphasized the need for distributed ISD teams to be agile, but observed that agility concepts were not being applied to manage such teams at TECHCOM (or other companies he was aware of), indicating *the need to bring agility concepts as applicable to this setting to the forefront*.

Furthermore, a TECHCOM manager reported that he/she had included some of the issues discussed in the interview in the Project Retrospectives document (prepared after the completion of a recent project), reflecting the fact that some of the tactics used or lessons learned about distributed ISD (including agility concerns) were not part of TECHCOM’s methodology and its standard team management practices (based on CMM, PSP, and TSP). Considering the above, we believe that *our work can play a valuable role in focusing practitioners’ attention on to agility concerns so as to enable more effective ISD in distributed settings*. Indeed, according to Astley and Zammuto (1992), new “linguistic constructs” and images in managerial language about a concept or phenomenon (in our case, on agility) are known to facilitate action in directions that were previously “unattended to or inconceivable” (p. 450)—we are hopeful that this work will provide “a linguistic medium of conceptual and symbolic discourse” (p. 457) on agility in distributed teams that could have such an impact in practice.

8.2. Conclusion

We would like to conclude by stating that past research has focused primarily on technological and economic considerations associated with distributed ISD and the offshoring “mega trend” (Robinson and Kalakota 2004). As this trend has matured and organizations have started viewing the distribution of ISD work from a strategic perspective, the need has been felt to reexamine the same phenomenon in light of other contemporary concerns, a critical one being that of “agility.” Our study takes a step in attempting to articulate core issues in this emerging arena of scholarship pertaining to today’s global and wired world within which ISD processes are embedded.

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